KNEXUS

Dataset Selection & Evaluation Strategy

Knowledge Network for EXploration and Unified Services

Major Project Review I | VII Semester | Academic Year 2026–2027

Prepared as a practical dataset recommendation for the KNEXUS project team.

1. Executive Recommendation

Use EnterpriseRAG-Bench as the primary public corpus for KNEXUS's organizational knowledge-retrieval and project-coordination use cases. Do not treat it as the sole dataset for the whole system. Add a small, purpose-built synthetic test set for role-based access control (RBAC), cross-tenant isolation, approval handling, tool execution, post-condition verification, and audit logging.

Core decision: EnterpriseRAG-Bench for enterprise knowledge; custom KNEXUS fixtures for security and governed actions.

2. What KNEXUS Needs to Evaluate

KNEXUS is more than a document chatbot. The project blueprint describes an organization-aware assistant that retrieves evidence from internal-style information, answers with citations, abstains when evidence is insufficient, enforces server-side permissions, and routes approved actions through a controlled workflow.

Requirement	What the data or tests must support
Company knowledge	Policies, procedures, documentation, project records, and organizational communications.
Project coordination	Questions that connect task records, discussions, emails, meeting notes, and project documents.
Grounded answers	Reference answers and identifiable supporting documents or passages.
Permission-aware retrieval	Tests showing that users can retrieve only the records they are authorized to access.
Controlled actions	Tests for input validation, authorization, approval, execution, verification, and audit events.

3. Primary Dataset: EnterpriseRAG-Bench

EnterpriseRAG-Bench is a benchmark built around simulated company information. The source material describes a fictional organization, Redwood Inference, and records across workplace systems. These are synthetic records, not evidence obtained from a real company's private systems.

Official resources:

- **GitHub repository:** https://github.com/onyx-dot-app/EnterpriseRAG-Bench
- **Hugging Face dataset:** https://huggingface.co/datasets/onyx-dot-app/EnterpriseRAG-Bench
- **Methodology:** https://github.com/onyx-dot-app/EnterpriseRAG-Bench/blob/main/methodology.md

3.1 Why it fits KNEXUS

The dataset represents multiple workplace sources, making it a closer fit for enterprise knowledge retrieval than a single-domain FAQ or isolated-document QA set. Questions may require connecting evidence across project-management records, internal conversations, emails, meeting transcripts, and shared documentation.

Information source	Approximate published records	Potential KNEXUS use
Slack	285,605	Discussions, decisions, status updates.
Gmail	121,390	Email threads and organizational communication.
Linear	35,308	Tasks, issues, and project work items.

Information source	Approximate published records	Potential KNEXUS use
Google Drive	25,108	Shared documents and project documentation.
HubSpot	15,017	Customer and sales records, if in scope.
Fireflies	10,173	Meeting transcripts and decisions.
GitHub	8,052	Development discussions and pull requests.
Jira	6,120	Tickets and issue tracking.
Confluence	5,189	Wikis, internal documentation, and runbooks.

Counts above are approximate published counts reported in the earlier dataset review. Recheck the current repository and dataset card before using them in a formal report, because dataset versions or published counts can change.

3.2 Evaluation question categories

Category	Count	Why it matters
Basic	175	Retrieve a straightforward answer from relevant evidence.
Semantic	125	Handle paraphrases and vocabulary differences.
Intra-document reasoning	40	Combine information across sections of one document.
Project-related	40	Connect records associated with a project or initiative.
Constrained	30	Disambiguate similar records or satisfy constraints.
Conflicting information	20	Recognize and handle inconsistent evidence.
Completeness	20	Retrieve all relevant evidence needed for an answer.
Miscellaneous	20	Handle less-structured or unusual cases.
High-level	10	Answer broader questions requiring synthesis.
Information not found	20	Abstain when the evidence does not establish an answer.
Total	500	Published category counts sum to 500.

Prioritize project-related, conflicting-information, and information-not-found questions because they exercise project coordination, evidence comparison, and abstention. Also include semantic and constrained questions so evaluation does not reduce to simple keyword matching.

4. Relationship to the Research Papers

Paper / research direction	What it contributes	Dataset implication for KNEXUS
Lewis et al., Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (2020)	Motivates combining a language model with retrieved external evidence.	A company-oriented corpus is more representative of KNEXUS's target use case than adopting Wikipedia merely because it appears in the paper's experiments.
Schick et al., Toolformer (2023)	Studies language models learning to call tools and construct API arguments.	Does not supply KNEXUS's company records, permissions, approval states, or audit scenarios; build controlled tool tests

Paper / research direction	What it contributes	Dataset implication for KNEXUS
		separately.
Zhao et al., AgentCDM (2025 preprint)	Studies collaborative decision-making and structured reasoning.	May be indirectly relevant to synthesizing or comparing evidence, but is not a substitute for an enterprise knowledge corpus.

Use the papers to justify design choices, not as an automatic instruction to reuse their experimental datasets. Verify bibliographic details and claims against the original publications before including them in the final academic report.

5. Recommended Corpus Size and Selection

Begin with approximately 1,000–2,000 carefully selected records as an implementation starting point, not as a benchmark requirement or a claim about measured performance. A smaller, coherent corpus will be easier to ingest, inspect, debug, and evaluate than the full collection.

- Select a connected set of project tasks, conversations, documents, emails, meeting notes, and internal documentation.
- Keep the records needed to answer selected evaluation questions, plus related records and plausible distractors.
- Avoid selecting only the exact answer-bearing documents; that would make retrieval artificially easy.
- Preserve original document IDs, source type, title, content, and relationships when possible.
- Use a direct importer if practical; otherwise convert selected records to supported formats while retaining source metadata.
- Check current repository licensing, access instructions, file formats, and data-card details before downloading or redistributing data.

Suggested source priorities

Priority	Source	Reason
1	Linear	Project tasks and work items.
2	Slack	Project discussions and updates.
3	Google Drive	Project documents and requirements.
4	Gmail	Cross-team communication.
5	Fireflies	Meeting discussions and decisions.
6	Confluence	Procedures and internal documentation.
7	GitHub / Jira	Development progress and technical records.

This is a suggested selection order, not a claim that these sources are universally most important.

6. Permission and Security Data Must Be Added Separately

The reviewed dataset description documents source and content fields, as well as question/reference fields. It does not establish that a ready-made user/role/tenant access-control annotation system is included. Do not assume that loading the corpus automatically evaluates RBAC or tenant isolation.

6.1 Suggested KNEXUS metadata

Field	Purpose
organization_id	Identifies the organization that owns a record.
source_type	Identifies the originating workplace system.
visibility	Represents organization-wide or restricted visibility.
allowed_roles	Specifies which roles may access the record.
project_id	Associates a record with a project where relevant.
document_id	Preserves the source dataset identifier.

These are proposed application fields, not claims about existing EnterpriseRAG-Bench fields. Apply authorization on the server and filter inaccessible evidence before it is passed to the language model. A frontend-only filter is not an adequate security boundary.

6.2 Required security test cases

- An authorized user can retrieve an allowed document.
- A user cannot retrieve a restricted document even when it closely matches the query.
- A user from one synthetic tenant cannot access another tenant's records.
- A permission change or revocation takes effect as designed.
- Retrieved prompt-injection content is treated as untrusted data rather than instructions to override system policy.
- The assistant abstains when the user's accessible records do not contain enough evidence.

7. Separate Synthetic Data for Tool Execution

EnterpriseRAG-Bench should not be treated as a complete action-lifecycle dataset. Create a small synthetic application database and a controlled local tool to test KNEXUS's action workflow.

Example: Website Migration project

- Create a fictional project with tasks, assigned users, deadlines, dependencies, and status.
- Add a few synthetic discussions or meeting notes that describe outstanding work.
- Define a tool such as create_project_task that writes only to the local test database.
- Define who may propose an action and who may approve it.
- Record the action proposal, approval decision, execution result, verification result, and audit event.

Test scenario	Expected behavior
Authorized user proposes a valid task	Validate the request and apply the configured approval policy.
Unauthorized user requests an operation	Reject the operation server-side.
Approval has expired	Do not execute the action.
Approved payload is modified	Reject a payload that no longer matches the approved proposal.
Task is created successfully	Verify the task exists and record the outcome.
Execution fails or the expected state is absent	Record failure or an inconclusive result; do not claim success.

8. Should External Internet Data Be Included?

For the initial MVP, keep unrestricted external web retrieval out of scope unless it is explicitly required. First demonstrate authorized internal-style retrieval, citations, abstention, and controlled actions.

If external retrieval becomes a later requirement, CRAG (Comprehensive RAG Benchmark) is a possible supplementary evaluation resource because it focuses on web-search results and mock knowledge-graph/API retrieval. It is not the preferred replacement for an internal enterprise corpus.

If external sources are added, label them separately, preserve provenance, and apply explicit trust/source policies. Do not silently present an external web result as an approved internal policy.

9. Alternatives Considered

Dataset	Main focus	Assessment for KNEXUS
EnterpriseRAG-Bench	Simulated enterprise records across workplace tools	Best primary fit among the reviewed options; requires separate security and action tests.
Doc2Dial	Document-grounded, goal-oriented dialogue	Useful for conversational retrieval, but narrower than a multi-source company workspace.
TechQA	Technical-support QA grounded in technical documents	Useful for technical support, but less representative of cross-functional project coordination.
CRAG	RAG with web-search content and mock knowledge-graph/API results	Useful as an optional external-information evaluation resource.

These alternatives are valid for their intended tasks; they are not selected as the primary corpus because they are less aligned with KNEXUS's internal enterprise coordination requirements.

10. Concrete Dataset and Evaluation Plan

Component	Recommended data	Purpose
Primary knowledge corpus	Selected EnterpriseRAG-Bench subset	Internal-style knowledge and project coordination.
Retrieval / answer evaluation	Benchmark questions, gold answers, expected document IDs	Measure retrieval and evidence-grounded answering.
Security evaluation	Custom synthetic roles, tenants, document permissions, and adversarial tests	Evaluate authorization, isolation, and abstention under permissions.
Action evaluation	Synthetic project records and a local sandbox tool	Evaluate validation, approval, execution, verification, and audit logging.

10.1 Evaluation metrics

Area	Possible measure	How to evaluate
Retrieval	Recall@k / Precision@k	Compare retrieved document IDs with expected supporting-document IDs.
Citation quality	Citation support rate	Check whether cited passages support the claims they are attached to.
Answer reliability	Answer correctness and abstention behavior	Compare answers with reference answers and inspect unsupported-answer cases.
Authorization	Unauthorized retrieval/action rate	Run allowed and denied access tests across synthetic roles and tenants.
Tool workflow	Validation, approval, execution, and verification success	Run positive and negative action scenarios.
Auditability	Required-event traceability	Verify each test action produces the required traceable audit records.

Area	Possible measure	How to evaluate
Performance	Latency and resource usage	Measure under a documented, repeatable test setup.

Do not report numerical results until experiments have been run. Reserve part of the question set for final evaluation and avoid repeatedly tuning the system against every test question.

11. Practical Implementation Notes

- A foundation model does not need to be trained from scratch for this project.
- Ingest the chosen records into the retrieval pipeline, preserve metadata, retrieve relevant evidence, and generate answers using the selected local model runtime.
- Keep original IDs and provenance so evaluation can map retrieved evidence to expected documents.
- Ensure the evaluation setup records model, embedding model, chunking strategy, retrieval settings, and prompt version.
- Use only data permitted by the dataset's current license and terms; do not upload confidential company information without authorization.
- A synthetic benchmark can demonstrate behavior on simulated records, but does not prove identical performance on real company data.

12. Final Verdict

Recommended primary dataset: EnterpriseRAG-Bench.

Recommended overall data strategy: EnterpriseRAG-Bench for company knowledge and project coordination, its reference questions for RAG evaluation, and custom synthetic KNEXUS fixtures for RBAC, cross-tenant isolation, approval, controlled tool execution, post-condition verification, and audit logging.

Do not begin with the full corpus. Start with a coherent subset of approximately 1,000–2,000 records, validate the ingestion pipeline, and expand only when justified by evaluation. Keep external web retrieval optional for the initial MVP.

13. References and Official Resources

1. **EnterpriseRAG-Bench GitHub repository:** https://github.com/onyx-dot-app/EnterpriseRAG-Bench
2. **EnterpriseRAG-Bench Hugging Face dataset:** https://huggingface.co/datasets/onyx-dot-app/EnterpriseRAG-Bench
3. **EnterpriseRAG-Bench methodology:** https://github.com/onyx-dot-app/EnterpriseRAG-Bench/blob/main/methodology.md
4. **Doc2Dial paper (ACL Anthology):** https://aclanthology.org/2020.emnlp-main.652/
5. **TechQA (IBM Research):** https://research.ibm.com/publications/the-techqa-dataset
6. **CRAG repository:** https://github.com/facebookresearch/CRAG
7. **RAG paper (arXiv):** https://arxiv.org/abs/2005.11401
8. **Toolformer paper (arXiv):** https://arxiv.org/abs/2302.04761
9. **AgentCDM paper (arXiv):** https://arxiv.org/abs/2508.11995

Verification note: dataset record counts and schema descriptions in this document reflect the research summary prepared for this recommendation. Before submitting the document as a formal academic report, confirm current counts, licenses, download formats, and paper metadata on the linked official pages.